

Vision-Driven House Price Estimation: A Hybrid Convolutional and Regression-Based Multimodal Learning Approach

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[Project GitHub Link](#)

Abstract—Traditional house price prediction models rely primarily on structured attributes (e.g., size, bedrooms/bathrooms, and location proxies) and often fail to capture visual cues such as architectural style, lighting, and property condition that influence perceived value. This paper investigates multimodal house price estimation by comparing three learning approaches: a tabular-only regression model trained on preprocessed housing attributes (including categorical location encoded appropriately), an image-only model that uses a convolutional neural network (CNN) to extract visual embeddings from property photos, and a hybrid model that fuses both modalities by concatenating tabular and CNN-derived feature representations followed by a regression head. All models are trained with mean squared error loss and evaluated using MSE and R^2 , enabling a direct comparison of unimodal versus multimodal performance.

Index Terms—Multimodal learning, Tabular Regression, Convolutional Neural Networks (CNN), Feature Fusion, Computer Vision

I. INTRODUCTION

Accurate residential property valuation is important for buyers and sellers, lenders, insurers, and online listing platforms. Classical approaches often follow the hedonic pricing framework, modeling sale price as a function of structural and locational attributes such as size, number of rooms, and neighborhood characteristics [1]. While modern machine learning models can improve predictive performance, many practical pipelines still rely primarily on structured tabular features and do not explicitly account for visual appearance.

In real estate, visual characteristics such as architectural style, maintenance level, and interior finishes can influence perceived value. This motivates incorporating computer vision representations derived from property photographs into price estimation. Prior work has shown that convolutional neural networks can extract useful visual signals for valuation, and that combining visual and non-visual attributes can improve performance [3]–[5].

In this paper, I compare three models under a shared experimental pipeline: (i) a tabular-only regressor trained on numerical and categorical housing features (including location encoded categorically), (ii) an image-only regressor that predicts price from property images using a CNN,



and (iii) a hybrid fusion model that concatenates tabular and visual embeddings before a regression head. Models are trained using mean squared error loss and evaluated using mean squared error (MSE) and the coefficient of determination (R^2) on a held-out test set.

So overall, this paper gives a: (i) an end-to-end reproducible pipeline for unimodal and multimodal price prediction, (ii) a controlled comparison between tabular-only, image-only, and fusion models, and (iii) an empirical analysis of the effect of incorporating property images alongside structured attributes.

II. LITERATURE SURVEY

A. Background

House price prediction is commonly framed as a supervised regression task: given an input representation of a property, the goal is to predict a continuous target (sale price). Classical real estate valuation often models price as a function of structural and neighborhood characteristics [1]. In modern settings, this formulation extends naturally to machine learning models that learn nonlinear mappings from features to prices.

Convolutional neural networks (CNNs) are a standard architecture for learning representations from images

[11], [12]. Although CNNs are widely used for image classification, they can be adapted for regression by replacing the final classification layer with a regression head that outputs a single value and training with a regression loss such as mean squared error [3]. In practice, CNN backbones such as ResNet [6] are frequently used as feature extractors, producing a fixed-length embedding that is mapped to the target via a small multilayer perceptron.

A common strategy for vision-based regression is transfer learning. CNNs pretrained on large-scale datasets such as ImageNet [7] provide generic visual features that can improve performance and reduce training time when labeled task-specific data is limited. This approach, fine-tuning a pretrained backbone and training a task-specific head, is a standard application of transfer learning in deep learning systems [8], [9] and has been adopted in vision-based housing price estimation [5].

B. Related Work

Vision-based real estate valuation has been studied using interior and exterior photographs as well as neighborhood imagery. Poursaeed et al. demonstrated that visual features extracted by CNNs can predict real estate prices and complement non-visual cues [3]. Law et al. incorporated street view and satellite imagery to estimate house prices, highlighting the predictive value of neighborhood appearance [4]. Nouriani and Lemke further explored combining interior, exterior, and satellite images for housing price estimation [5]. These works motivate a comparative evaluation of unimodal tabular models, unimodal vision models, and multimodal fusion models for price prediction.

Multimodal machine learning provides general principles for integrating heterogeneous modalities through strategies such as late fusion, where separate encoders are combined via concatenation before a prediction head [2]. Following this paradigm, this study evaluates whether fusing structured housing attributes with CNN-derived image embeddings yields improvements over unimodal baselines in a controlled experimental setting.

III. METHODOLOGY

A. Dataset

The *House Prices and Images - SoCal* dataset published on Kaggle [10] has been used. The dataset provides paired inputs for multimodal regression, consisting of (i) a CSV file of structured housing attributes and (ii) a corresponding set of exterior house images. Each tabular record contains an image identifier that links to its associated image file. The prediction target is the sale price (USD).

a) Outlier filtering: In the raw dataset, extremely high-priced homes above \$2,000,000 occur very rarely. Because these cases have insufficient representation (only 6 samples in total in the dataset) and lead to unstable training and unreliable per-range evaluation, the samples with price

> \$2,000,000 are removed prior to splitting. After this filtering step, the dataset contains $N = 15,468$ samples.

b) Price-bin distribution: To characterize the remaining data and support stratified splitting, discretized sale prices into three bins: \$0–500k, \$500k–1M, and \$1M–2M. The overall distribution after filtering is: 5,189 (33.55%) in \$0–500k, 8,186 (52.92%) in \$500k–1M, and 2,093 (13.53%) in \$1M–2M.

Price range	Train	Val/Test	Total
\$0–500k	4,151 (33.53%)	1,038 (33.54%)	5,189 (33.53%)
\$500k–1M	6,549 (52.90%)	1,637 (52.89%)	8,186 (52.90%)
\$1M–2M	1,679 (13.56%)	420 (13.57%)	2,099 (13.56%)
Total	12,376	3,092	15,468

TABLE I: Price-bin distribution after removing prices above \$2M and performing an 80/20 stratified split (seed=42). Percentages are within each split

B. Data Split and Stratification

The filtered dataset is split into 80% training and 20% test/validation using a fixed random seed of 42. To reduce distribution shift across splits, I stratify the split by the discrete price bins in Table I, ensuring each split preserves approximately the same proportion of low-, mid-, and high-priced homes. This stratification helps promote *balanced evaluation across price ranges*, so that reported performance is not dominated by the most common price band and predictive accuracy can be assessed more consistently across the full range of observed prices.

C. Tabular Preprocessing

Let X denote the structured input features and y denote the sale price. I first remove non-predictive identifiers and free-text fields (e.g., image identifiers and street strings) and retain numeric attributes such as bedrooms, bathrooms, and square footage, together with a categorical location feature (city). For numeric columns, missing values are imputed using the median and features are standardized using z-score normalization. For categorical columns, missing values are imputed with a constant token and features are one-hot encoded; unseen categories at test time are ignored to ensure a consistent feature dimension between training and evaluation.

Because price is right-skewed, I apply a $\log(1 + y)$ transform to the training target to improve numerical stability and reduce the influence of extreme values. Model outputs are trained and predicted in log space.

D. Image Preprocessing

Each property image is loaded and converted to RGB, then resized to 224×224 pixels to match the input resolution expected by the ResNet-18 backbone. Images are

converted to tensors and normalized using ImageNet channel statistics (mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]). During data loading, if an image file is missing or cannot be decoded, a blank placeholder image is substituted to avoid interrupting training and evaluation.

E. Models

- **Tabular-only model [Figure 1]:** A multilayer perceptron (MLP) regressor has been used that maps the preprocessed tabular feature vector to a single price prediction. The network contains two hidden layers with widths 128 and 64. Each hidden layer applies a linear transform followed by batch normalization, ReLU activation, and dropout ($p = 0.1$). A final linear layer outputs a single scalar.
- **Image-only model [Figure 2]:** For image-based prediction, ResNet-18 backbone pretrained on ImageNet is used. The final classification layer is replaced with an identity mapping to obtain a fixed-length image embedding (512 dimensions for ResNet-18), which is then passed through a regression head consisting of Linear (512→128) → ReLU → Linear(128→1). The backbone is fine-tuned during training.
- **Hybrid model (late fusion) [Figure 3]:** The hybrid model consists of two branches: a tabular branch (the MLP feature extractor up to the 64-dimensional hidden representation) and an image branch (the 512-dimensional ResNet-18 embedding). The two embeddings are concatenated to form a 576-dimensional fused representation, which is passed through a fusion head: Linear(576→128) → BatchNorm → ReLU → Dropout ($p = 0.2$) → Linear (128→64) → ReLU → Linear (64→1). This late-fusion design enables the model to learn complementary information from structured attributes and visual appearance in a single end-to-end predictor.

F. Training Configuration

All models are trained to minimize mean squared error (MSE) using the Adam optimizer with mini-batch gradient-based optimization. Each epoch iterates over the training set in batches of size 32, and model parameters are updated once per mini-batch using backpropagation. Training is performed for 20 epochs (i.e., a fixed training budget with no early stopping). The tabular-only model is trained with learning rate 0.01, while the image-only model is trained with learning rate 0.0001 to support stable fine-tuning of the convolutional backbone.

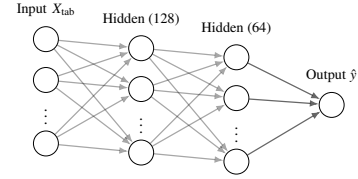
For the hybrid model, differential learning rates are used via optimizer parameter groups: the CNN branch is fine-tuned with learning rate 0.0001, while the tabular branch and fusion head are trained with learning rate 0.01. To improve training stability, gradients are clipped with maximum norm 1.0 prior to each optimizer step.

Additionally, a ReduceLROnPlateau schedule on is applied on the validation loss (factor=0.1, patience=2 epochs), reducing the learning rate when validation performance plateaus.

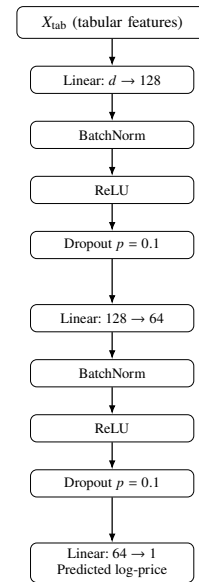
G. Evaluation Metrics

Predictive performance is evaluated using mean squared error (MSE) and the coefficient of determination (R^2). MSE directly measures average squared deviation between predicted and true prices, while R^2 summarizes the fraction of variance in the target explained by the model relative to a mean-price baseline. Since house prices are strongly right-skewed, training is performed on the transformed target $y' = \log(1 + y)$ to stabilize optimization. Accordingly, MSE and R^2 is also reported in log space, which reduces the influence of extreme prices and reflects relative (multiplicative) errors more consistently across the price range.

Fig. 1: Tabular-only model diagrams: (a) neuron-level schematic, (b) computation graph view

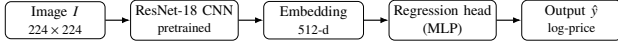


(a) MLP structure (layer widths shown; dots indicate additional neurons).

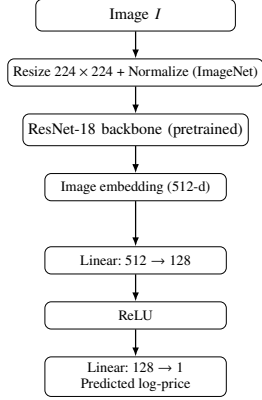


(b) Forward computation with normalization and dropout used in the implementation.

Fig. 2: Image-only model diagrams: (a) schematic view, (b) computation graph view.

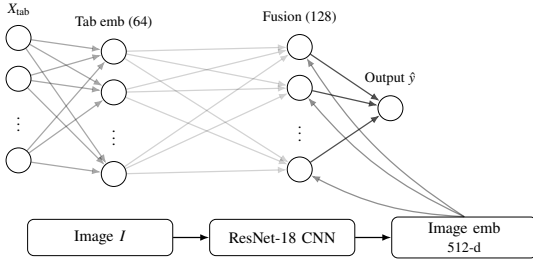


(a) High-level schematic (CNN backbone + regression head).

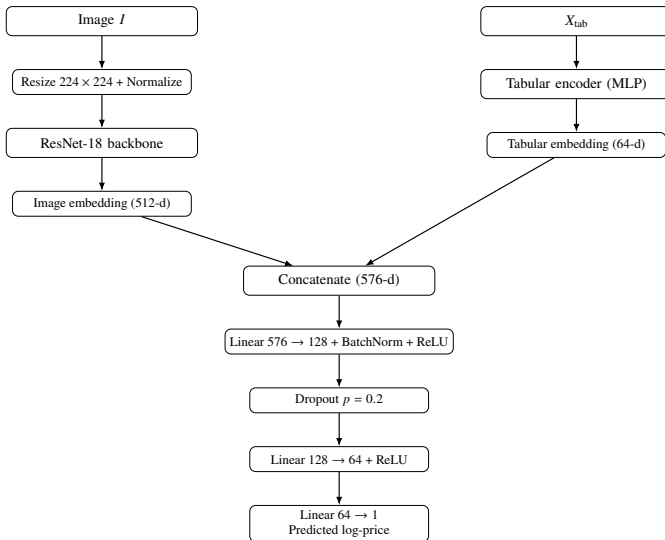


(b) Forward computation used in the implementation.

Fig. 3: Hybrid (late-fusion) model diagrams: (a) schematic view, (b) computation graph view.



(a) Schematic view (tabular branch + image branch + fusion).



(b) Forward computation used in the implementation (late fusion).

IV. RESULTS

This section reports experimental results for the three compared approaches: tabular-only, image-only, and hybrid (late fusion). All metrics and plots are computed in log space (targets are $\log(1 + y)$) to match the training objective and to reduce the influence of heavy right-skew in sale prices. Unless stated otherwise, results are reported on the held-out 20% split produced via stratification over the three price bins (\$0–500k, \$500k–1M, \$1M–2M) after removing listings above \$2M.

A. Training dynamics

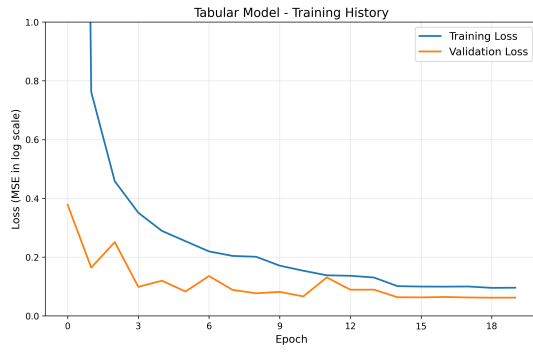
Figure 4 reports training and validation loss (MSE in log space) over 20 epochs. The three models exhibit noticeably different optimization and generalization behavior.

The **tabular-only model** (Fig. 4a) shows a steady decrease in training loss after an initially high first-epoch value, while validation loss drops sharply within the first few epochs and then stabilizes at a low level with minor fluctuations. The small train–validation gap and the relatively flat validation curve after convergence suggest good generalization under the chosen preprocessing and learning rate.

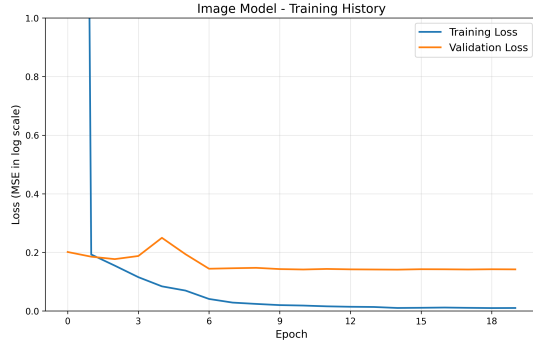
The **image-only model** (Fig. 4b) converges very quickly in training, with the training loss approaching near-zero after the early epochs. However, the validation loss improves only modestly and then plateaus (with an early bump around epochs 3–4). This creates a clear and persistent generalization gap, indicating overfitting: the CNN regressor fits the training images well but does not translate proportionally to improved validation performance, consistent with imagery alone being an incomplete proxy for sale price.

The **hybrid model** (Fig. 4c) exhibits fast convergence similar to the image model in the first few epochs, followed by continued gradual improvement. Its validation loss decreases substantially after the initial epochs and stabilizes at a lower level than the tabular-only baseline, with only small oscillations thereafter. Overall, the hybrid training curves suggest that adding visual embeddings to structured attributes improves generalization compared to the image-only model while retaining stable optimization, supporting the benefit of multimodal fusion in this setting.

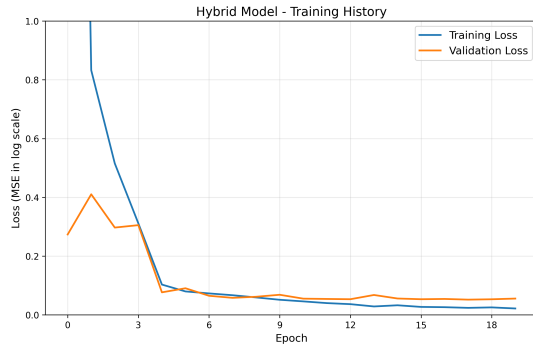
Additionally, Figure 5 directly compares validation loss across models, highlighting the consistently lower log-MSE achieved by the hybrid approach after early training.



(a) Tabular-only.



(b) Image-only.



(c) Hybrid (late fusion).

Fig. 4: Training histories (log-MSE) for tabular-only, image-only, and hybrid models.

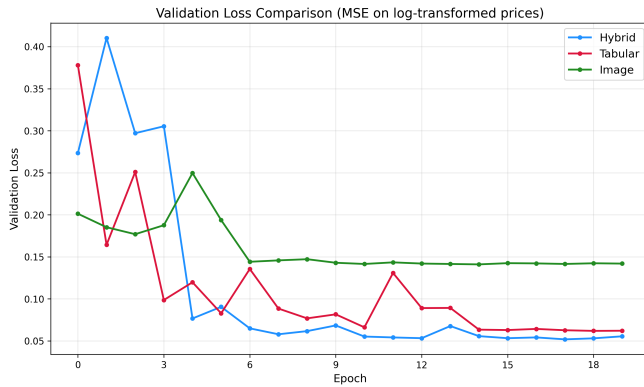
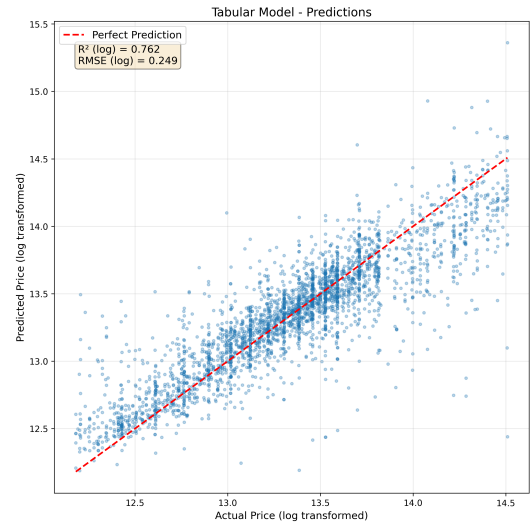
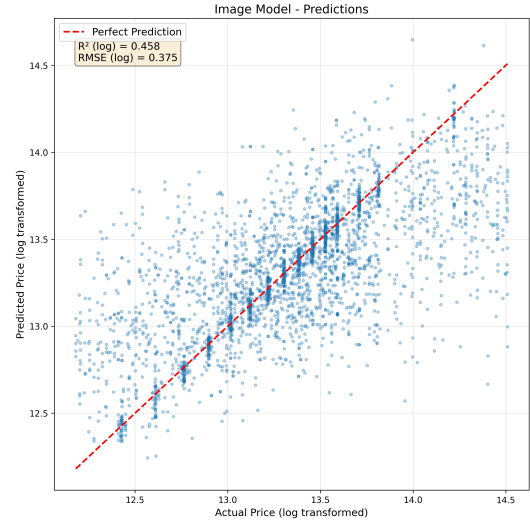


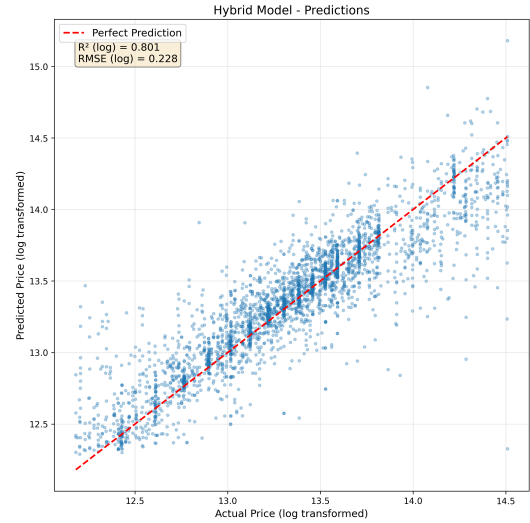
Fig. 5: Validation loss comparison (log-MSE) across tabular-only, image-only, and hybrid models over 20 epochs.



(a) Tabular-only.



(b) Image-only.



(c) Hybrid (late fusion).

Fig. 6: Predicted vs. actual log-prices. Points closer to the dashed identity line indicate better calibration.

B. Predicted vs. Actual Behavior

Figure 6 visualizes predicted vs. actual log-prices for each model (perfect predictions lie on the diagonal).

The image-only model produces the widest dispersion around the diagonal and exhibits noticeable bias at the upper end of the price range, where predictions tend to fall below the ideal line. This indicates limited ability to infer high-value properties from exterior imagery alone and aligns with its lower $R^2_{\log}$ and higher $RMSE_{\log}$.

The tabular-only model yields a substantially tighter cloud and stronger alignment with the diagonal, confirming that structured attributes and categorical city information capture much of the price signal. Nonetheless, some spread remains, particularly for higher-priced homes, indicating residual uncertainty not fully explained by tabular variables.

The hybrid model provides the closest alignment to the diagonal overall and the tightest concentration across most of the range. Its improvement over tabular-only (0.801 vs. 0.762 in $R^2_{\log}$) suggests that visual cues add incremental predictive value when fused with structured features, reducing error and improving explained variance.

C. Residual analysis

Residual diagnostics in Figure 8 highlight both random error spread and systematic bias. The tabular-only model produces residuals that are tightly centered around zero with a relatively sharp histogram peak, indicating low global bias; however, the residual scatter shows mild heteroscedasticity (slightly larger spread in the mid/high predicted range) and a few notable outliers, consistent with occasional hard-to-price listings. The image-only model exhibits the broadest residual distribution and a visibly structured residual pattern across predicted values (i.e., the cloud is not purely random around zero), suggesting systematic under/over-estimation for certain portions of the price range and weaker calibration when relying on exterior imagery alone. In contrast, the hybrid model yields the tightest residual band and the most concentrated histogram around zero, indicating reduced error variance and fewer large deviations overall. Taken together, these residual plots support the conclusion that late fusion improves robustness by compensating for modality-specific blind spots (missing structured factors in images, and missing visual curb-appeal cues in tabular features).

D. Overall metric comparison

Table II summarizes test performance for all three models using MSE, RMSE, and R^2 in both log space (training space) and original-dollar space (after inverting $\log(1 + y)$). As visualized in Figure 9 and Figure 10, the hybrid late-fusion model achieves the best overall fit, attaining the highest coefficient of determination in log space ($R^2_{\log} = 0.801$) and the lowest log-RMSE ($RMSE_{\log} = 0.228$). The tabular-only baseline remains competitive ($R^2_{\log} = 0.762$, $RMSE_{\log} = 0.249$), reflecting the strong predictive signal in structured

TABLE II: Validation-set performance for tabular-only, image-only, and hybrid (late fusion) models.

Model	$MSE_{\log}$	$RMSE_{\log}$	$R^2_{\log}$	R^2
Tabular	0.0619	0.2487	0.7620	0.6787
Image	0.1410	0.3755	0.4576	0.3756
Hybrid	0.0518	0.2276	0.8008	0.7288

attributes and location. In contrast, the image-only model underperforms substantially ($R^2_{\log} = 0.458$, $RMSE_{\log} = 0.375$), indicating that exterior imagery alone does not capture many key determinants of sale price.

E. Accuracy within relative error bands

To provide a more interpretable notion of practical utility, the fraction of predictions whose absolute percentage error in the original dollar space falls within preset tolerance bands (after inverting the $\log(1 + y)$ transform) is also reported. Figure 7 compares accuracy at $\pm 5\%$, $\pm 10\%$, $\pm 15\%$, and $\pm 20\%$.

Across all thresholds, the hybrid model achieves the highest coverage, with 30% of predictions within $\pm 5\%$, 51% within $\pm 10\%$, 63% within $\pm 15\%$, and 73% within $\pm 20\%$. The tabular-only model follows closely (25%, 44%, 60%, and 71%, respectively), indicating that structured features capture much of the price signal but benefit from the additional visual cues contributed by the image branch. The image-only model lags at each band (27%, 41%, 49%, and 55%), reinforcing that exterior imagery alone is insufficient to reliably infer many determinants of sale price. Notably, the hybrid model’s advantage is most pronounced at moderate tolerances (e.g., +7 points at $\pm 10\%$ and +14 points at $\pm 15\%$ versus image-only), suggesting fusion substantially increases the fraction of near-correct predictions rather than only improving average error.

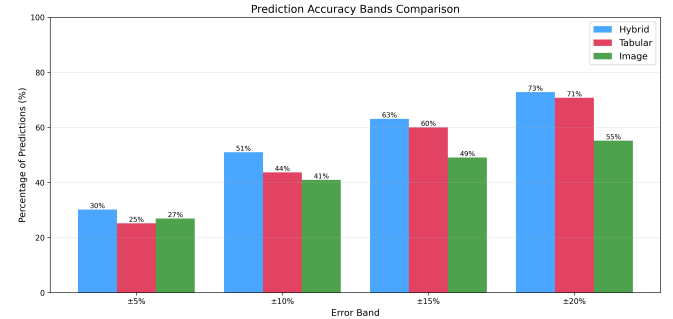
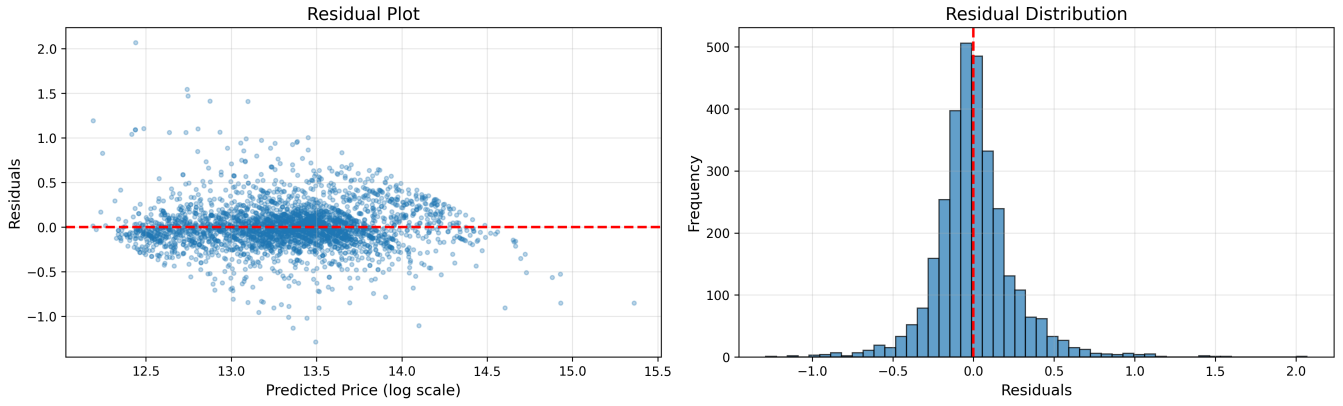


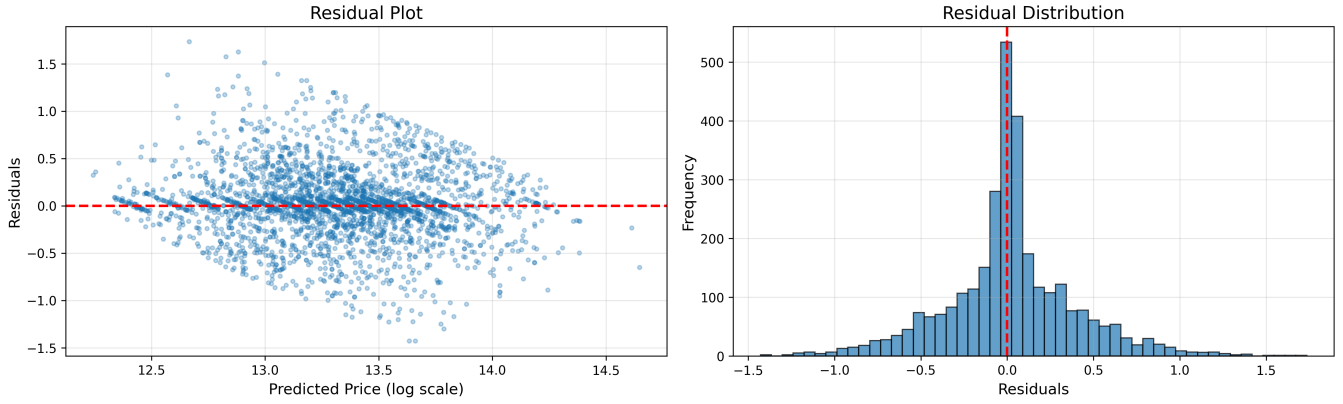
Fig. 7: Accuracy band comparison across models (percentage of predictions within each relative error threshold, computed in original dollar space).

Tabular Model - Residual Analysis



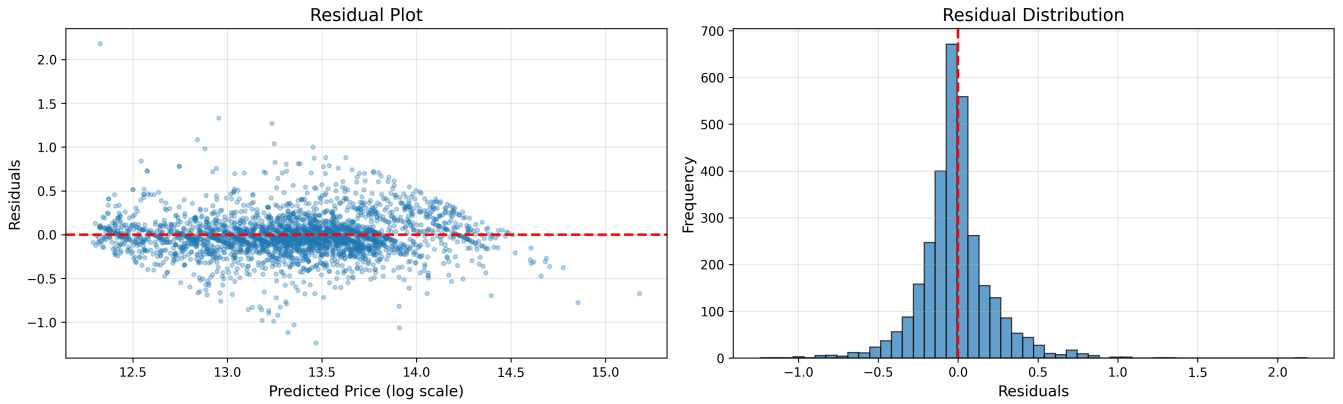
(a) Tabular-only.

Image Model - Residual Analysis



(b) Image-only.

Hybrid Model - Residual Analysis



(c) Hybrid (late fusion).

Fig. 8: Residual analysis in log space (scatter and histogram). Tighter residual bands indicate lower error dispersion.

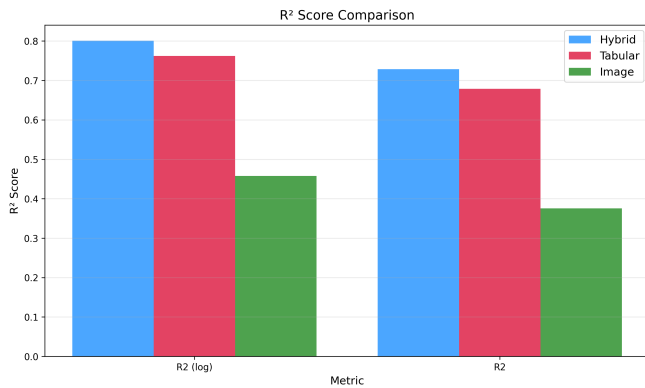


Fig. 9: R^2 comparison across models in log space and original scale. Higher is better.

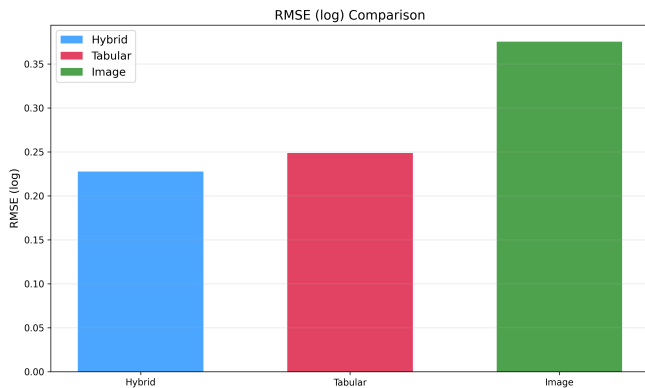


Fig. 10: $RMSE_{\log}$ comparison across models. Lower is better.

F. Performance across price ranges

Generalization is also evaluated across the three sale-price bins used for stratified splitting (\$0–500k, \$500k–1M, \$1M–2M). Because the image-only model produced strongly negative R^2 values in nearly every bin (indicating performance worse than predicting the within-bin mean), it has been omitted from the per-bin R^2 visualization and focus on the tabular and hybrid models. Figure 11 reports R^2 computed in log space within each bin, along with the number of test samples per bin.

Across the two lower ranges, both models achieve positive R^2 , with the hybrid model slightly higher and more consistent, suggesting that the fused representation provides a modest but reliable benefit when sufficient training support exists. In the \$500k–\$1M bin, the tabular model dips slightly below zero while the hybrid remains positive, implying that multimodal fusion better captures residual price drivers not fully explained by structured attributes alone.

In the \$1M–\$2M bin, both models yield substantially negative R^2 (hybrid less negative than tabular). This outcome should be interpreted cautiously. A negative R^2 does *not* necessarily imply extremely large absolute errors; rather, it indicates that, for that subset, the model’s squared

error exceeds the error of a trivial baseline that predicts the mean target for the bin. This can occur when (i) the bin has fewer samples (here, $n \approx 419$), increasing estimator variance; (ii) the within-bin target variance is relatively small in log space, so moderate errors can dominate the R^2 denominator; and/or (iii) the \$1M–\$2M segment contains heavier-tailed, idiosyncratic listings whose value depends on unobserved factors (renovations, interior finishes, lot characteristics, micro-neighborhood effects) not fully captured by the available tabular and street-view inputs. For this reason, per-bin RMSE (or median absolute percentage error) is often more stable than R^2 for small or noisy subpopulations; nevertheless, the hybrid model’s consistently less-negative R^2 suggests it degrades more gracefully in the highest price range.

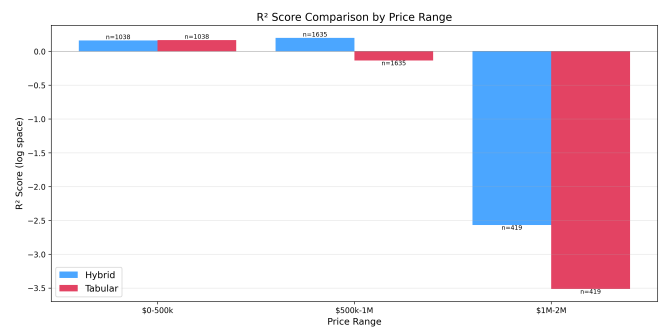


Fig. 11: Per-bin R^2 (log space) for hybrid vs. tabular models across price ranges. Negative R^2 indicates worse performance than predicting the within-bin mean.

V. DISCUSSION

All the experiments indicate that combining structured housing attributes with visual features yields the strongest performance, with the hybrid late-fusion model outperforming both unimodal baselines in $R^2_{\log}$ and error trends. This suggests that images contribute complementary information beyond the tabular variables, while the tabular modality supplies the dominant predictive signal.

A. Why the hybrid model improves over tabular-only

The tabular-only baseline achieves strong performance, which is expected because key price drivers (e.g., size proxies and location encoded via city) are explicit in the structured inputs. However, the hybrid model consistently improves on this baseline, implying that property photos contain residual information not fully captured by the tabular fields. Qualitatively, this extra signal likely corresponds to visual cues associated with perceived quality and condition (e.g., architectural style, renovation level, exterior materials, curb appeal), which can shift prices even among homes with similar recorded attributes. The hybrid model can exploit this by learning an image embedding that refines predictions in ambiguous regions of feature space.

B. Why the image-only model underperforms

The image-only model exhibits substantially lower predictive strength and shows a larger generalization gap between training and validation loss. This is consistent with two limitations: (i) exterior images alone are an incomplete representation of a home’s full value (interior condition, lot features, and neighborhood effects are only indirectly visible), and (ii) the dataset size and label noise make purely visual regression difficult. Even with transfer learning, the mapping from an exterior photo to sale price is weakly identifiable because many visually similar homes can differ greatly in location and lot-level attributes. As a result, the image-only model tends to regress toward average prices and produces a wider scatter around the $y = x$ line.

C. Training dynamics and generalization

Training curves reinforce the above interpretation. The tabular model converges smoothly and remains relatively stable, whereas the image-only model reduces training loss quickly but fails to achieve comparable validation improvement, indicating overfitting. The hybrid model converges quickly and sustains lower validation loss than either unimodal baseline, suggesting that tabular inputs provide a strong inductive bias while the image branch adds useful refinement without dominating optimization. This behavior is also consistent with using differential learning rates, where the CNN fine-tunes cautiously while the tabular and fusion heads adapt more rapidly.

D. Residual structure and error characteristics

Residual plots show that errors are not uniform across the target range: dispersion increases for higher-priced homes, which is expected in real estate valuation due to greater heterogeneity at the upper end of the market. Even in log space, luxury properties can differ substantially due to unobserved factors (micro-location, interior finishes, lot quality, views), which are not fully represented in a single exterior image and a limited set of structured attributes. The hybrid model reduces overall residual variance relative to the unimodal models, but heteroscedasticity remains, suggesting that future work should consider range-aware losses or evaluation by price bands.

E. Impact of stratified splitting and outlier handling

Initial per-range analyses highlighted that extreme price ranges (e.g., \$2M+) had too few samples to support stable training or reliable range-wise metrics. Removing these rare outliers and stratifying the train/test split by price bins improved comparability across models by ensuring that all splits contained similar proportions of low-, mid-, and high-priced homes. This reduces the chance that reported performance is dominated by the majority class (mid-priced homes) or distorted by a small number of extreme cases. Stratification therefore supports a more trustworthy

assessment of whether multimodal fusion offers benefits across the observed price spectrum.

F. Limitations

Several limitations constrain interpretation of the results. First, the model uses only exterior images; interior images and additional contextual imagery (e.g., street-view or satellite) could encode neighborhood and condition more directly. Second, labels may reflect temporal market fluctuations not modeled explicitly, introducing noise that may disproportionately affect the image-only approach. Third, the feature set may omit important structured predictors (e.g., school district, year built, renovation indicators), which can limit ceiling performance and create residual error even for the hybrid model.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This work evaluated multimodal house price estimation on the House Prices and Images – SoCal dataset by comparing three learning approaches: a tabular-only regressor trained on structured attributes (including categorical city), an image-only CNN regressor trained on exterior photos, and a hybrid late-fusion model that combines tabular and CNN embeddings. Across the held-out stratified split (after filtering listings above \$2M), the hybrid model achieved the strongest overall performance, improving over the tabular baseline and substantially outperforming the image-only model in both log-space and original-scale metrics. Training curves and error diagnostics additionally showed that the image-only model exhibits a larger train-validation gap, while fusing modalities yields more stable generalization. Overall, the results indicate that structured attributes capture most of the price signal, while visual cues provide complementary information that improves prediction when integrated through late fusion.

B. Future Work

Several extensions could strengthen both performance and experimental rigor (Figure 12:)

a) *Richer visual and multimodal modeling:* The current image branch uses a single exterior image and a standard CNN backbone. Future work could incorporate multiple views per property (exterior + interior), use attention-based aggregation over multiple photos, or adopt stronger pretrained vision encoders (e.g., larger CNNs or vision transformers) with appropriate regularization.

b) *Improved fusion strategies:* Late fusion via concatenation is simple and effective, but more expressive fusion mechanisms may better capture cross-modal interactions. Candidates include gated fusion, feature-wise modulation (FiLM), cross-attention, or mixture-of-experts designs that adaptively weight modalities based on input

quality.

c) Loss functions and target scaling: Although $\log(1 + y)$ stabilizes training, residual patterns suggest remaining heteroscedasticity. Future work could test robust losses (Huber, quantile loss) or explicitly model uncertainty (heteroscedastic regression) to better handle varying noise across the price range.

d) Stronger evaluation of tail behavior: Per-bin results show that high-price regimes are challenging, partly due to fewer samples and higher variance. Future work could (i) collect more high-end examples, (ii) evaluate with additional tail-robust metrics (median absolute percentage error, quantile-based metrics), and (iii) report confidence intervals via bootstrapping to quantify uncertainty in small subpopulations.

e) Feature enrichment for tabular inputs: The tabular model could benefit from additional location and neighborhood context (ZIP code granularity, geocoordinates, school scores, crime rates, walkability), which may reduce residual error attributed to micro-neighborhood effects not captured by city alone.

f) Generalization and replication: To assess robustness, future work could evaluate across regions (cross-dataset generalization), perform repeated runs over multiple random seeds, and report variability. Releasing preprocessing scripts and fixed splits would further support reproducibility and fair comparison across modeling choices.

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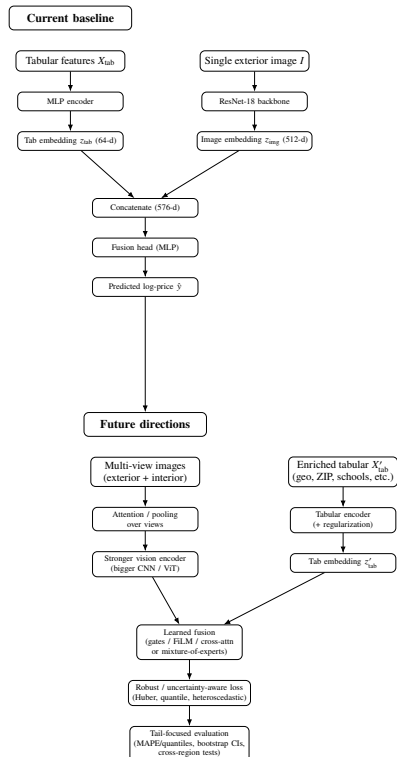


Fig. 12: Roadmap diagram summarizing future work